

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$B_{\mu\nu}$	H	$W_{\mu\nu}$	$H \ell$
T3-d3-d5-F4	0	3	0	5	1	4	$\frac{1}{2}$	$ \begin{aligned} & +C_{H2}H_{a_1}H_{a_1}^\dagger \\ & -H_{a_1}\mu^2H_{a_1}^\dagger \\ & +\frac{3\text{ImbdaHInv}2^2hH_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\ & +\frac{15\text{ImbdaHInv}1^2hH_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{14\epsilon} \\ & +\frac{95\text{ImbdaHInv}2^2hH_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{42\epsilon} \\ & +\frac{10\sqrt{\frac{2}{3}}\text{ImbdaHInv}1\text{ImbdaHInv}2H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{7\epsilon} \\ & -\frac{17\hbar B_{\mu_1\mu_2}^2}{12\epsilon} \\ & -\hbar\log\left(\frac{\mu\text{bar}^2}{M_F^2}\right)B_{\mu_1\mu_2}^2 \\ & -\frac{5}{12}\hbar\log\left(\frac{\mu\text{bar}^2}{M_{S_2}^2}\right)B_{\mu_1\mu_2}^2 \end{aligned} $	$ \begin{aligned} & +C_{H2}H_{a_1}H_{a_1}^\dagger \\ & -H_{a_1}\mu^2H_{a_1}^\dagger \\ & +\frac{3\text{ImbdaHInv}2^2hH_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\ & +\frac{15\text{ImbdaHInv}1^2hH_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{14\epsilon} \\ & +\frac{95\text{ImbdaHInv}2^2hH_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{42\epsilon} \\ & +\frac{10\sqrt{\frac{2}{3}}\text{ImbdaHInv}1\text{ImbdaHInv}2H_{a_1}H_{a_2}H_{a_1}^\dagger H_{a_2}^\dagger}{7\epsilon} \\ & +\frac{2\sqrt{\frac{2}{3}}\hbar\lambda\overline{\lambda_{T3}}H_{a_1}H_{a_2}\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\ & +\frac{2\sqrt{\frac{2}{3}}\hbar\lambda\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2-M_{S_2}^2} \\ & +\frac{2\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\mu\text{bar}^2}{M_{S_1}^2}\right)\lambda_{T3}H_{a_1}H_{a_2}M_{S_1}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2-M_{S_2}^2} \\ & -\frac{2\sqrt{\frac{2}{3}}\hbar\lambda\overline{\lambda_{T3}}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2-M_{S_2}^2} \\ & +\frac{2\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\mu\text{bar}^2}{M_{S_2}^2}\right)\lambda_{T3}H_{a_1}H_{a_2}M_{S_2}^2\lambda_{T3}H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2-M_{S_2}^2} \end{aligned} $	$ \begin{aligned} & -\frac{6\hbar\left(W_{\mu_1\mu_2}^{A_1}\right)^2}{\epsilon} \\ & -5\hbar\log\left(\frac{\mu\text{bar}^2}{M_F^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2 \\ & -\frac{1}{6}\hbar\log\left(\frac{\mu\text{bar}^2}{M_{S_1}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2 \\ & -\frac{5}{6}\hbar\log\left(\frac{\mu\text{bar}^2}{M_{S_2}^2}\right)\left(W_{\mu_1\mu_2}^{A_1}\right)^2 \end{aligned} $	$ \begin{aligned} & \frac{2\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\mu\text{bar}^2}{M_F^2}\right)\overline{\lambda_{T1}\tau_1}\overline{\lambda_{T2}\tau_2}\lambda_{T4}\tau_4}{\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}C P_2\ell_{a_2\tau_1}H_{a_1}H_{a_3}M_{S_2}^2\lambda_{T1}\tau_1a_{2\tau_2}\ell_{a_3\tau_3}M_F^2 \\ & +\frac{2\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\mu\text{bar}^2}{M_F^2}\right)\overline{\lambda_{T1}\tau_1}\overline{\lambda_{T2}\tau_2}\lambda_{T4}\tau_4}{\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}C P_2\ell_{a_2\tau_1}H_{a_1}H_{a_3}M_{S_1}^2\lambda_{T1}\tau_1a_{2\tau_2}\ell_{a_3\tau_3}M_F^2 \\ & +\frac{2\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\mu\text{bar}^2}{M_F^2}\right)\overline{\lambda_{T1}\tau_1}\ell_{a_2\tau_1}C P_2\ell_{a_3\tau_2}M_{S_1}^2M_{S_2}^2\lambda_{T1}\tau_1\overline{\lambda_{T2}\tau_2}\ell_{a_3\tau_3}a_{2\tau_4}H_{a_1}^\dagger H_{a_2}^\dagger M_F^2}{\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)} \\ & +\frac{2\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\mu\text{bar}^2}{M_F^2}\right)\overline{\lambda_{T1}\tau_1}\ell_{a_2\tau_1}C P_2\ell_{a_3\tau_2}M_{S_1}^2M_{S_2}^2\lambda_{T1}\tau_1\overline{\lambda_{T2}\tau_2}\ell_{a_3\tau_3}a_{2\tau_4}H_{a_1}^\dagger H_{a_2}^\dagger M_F^2}{\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)} \\ & +\frac{2\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\mu\text{bar}^2}{M_{S_1}^2}\right)\overline{\lambda_{T1}\tau_1}\overline{\lambda_{T2}\tau_2}\lambda_{T4}\tau_4}{\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}C P_2\ell_{a_2\tau_1}H_{a_1}H_{a_3}M_{S_1}^2M_{S_2}^2\lambda_{T1}\tau_1a_{2\tau_2}\ell_{a_3\tau_3}M_F \\ & +\frac{2\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\mu\text{bar}^2}{M_{S_1}^2}\right)\overline{\lambda_{T1}\tau_1}\overline{\lambda_{T2}\tau_2}\lambda_{T4}\tau_4}{\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)}C P_2\ell_{a_2\tau_1}H_{a_1}H_{a_3}M_{S_1}^2M_{S_2}^2\lambda_{T1}\tau_1a_{2\tau_2}\ell_{a_3\tau_3}M_F \\ & +\frac{2\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\mu\text{bar}^2}{M_{S_1}^2}\right)\overline{\lambda_{T1}\tau_1}\ell_{a_2\tau_1}C P_2\ell_{a_3\tau_2}M_{S_1}^2M_{S_2}^2\lambda_{T1}\tau_1\overline{\lambda_{T2}\tau_2}\ell_{a_3\tau_3}a_{2\tau_4}H_{a_1}^\dagger H_{a_2}^\dagger M_F}{\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)} \\ & +\frac{2\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\mu\text{bar}^2}{M_{S_2}^2}\right)\overline{\lambda_{T1}\tau_1}\ell_{a_2\tau_1}C P_2\ell_{a_3\tau_2}M_{S_1}^2M_{S_2}^2\lambda_{T1}\tau_1\overline{\lambda_{T2}\tau_2}\ell_{a_3\tau_3}a_{2\tau_4}H_{a_1}^\dagger H_{a_2}^\dagger M_F}{\left(M_F^2-M_{S_1}^2\right)\left(M_F^2-M_{S_2}^2\right)\left(M_{S_1}^2-M_{S_2}^2\right)} \end{aligned} $

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	e
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